

MAT2540, Classwork19, Spring2026

11.1 Introduction to Trees

1. **Definition.** Trees: A tree is a connected undirected graph with no simple circuits

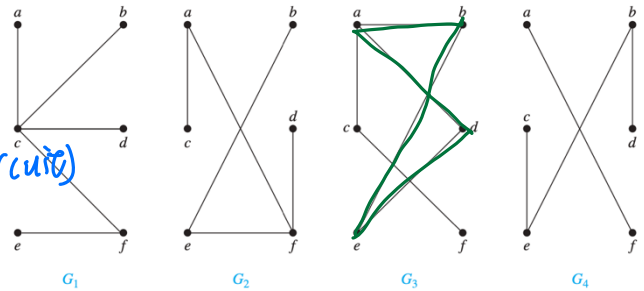
2. Examples. Check if each given graph is a tree.

G_1 : Yes, it is a tree

G_2 : Yes, it's a tree

G_3 : NO, it's not a tree (there is a circuit)

G_4 : NO, it's not a tree since it is not connected



3. **Definition.** Forest:

Graphs containing no simple circuit and not connected are called forests and have the property that each of their connected components is a tree. for example, G_4 in 2.

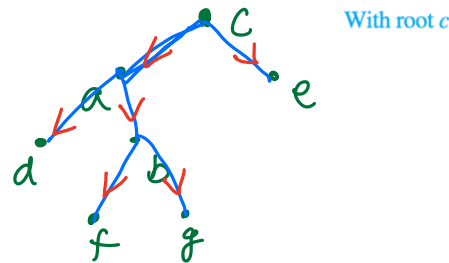
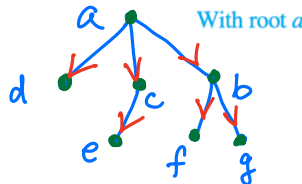
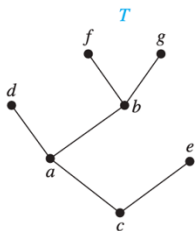
4. **Theorem:** An undirected graph is a tree if and only if there is a unique simple path between any two of its vertices.

5. **Definition.** A Rooted Tree:

A rooted tree is a tree in which one vertex has been designated as a root and every edge is directed away from the root.

We can change an unrooted tree into a rooted tree by choosing any vertex as a root

6. To get a rooted tree from an unrooted tree, different choices of the root produce different rooted trees.



7. The Terminology For Trees. Suppose that T is a rooted tree.

Parent: If v is a vertex in T other than the root, the *parent* of v is the unique vertex u such that there is a directed edge from u to v

Child: When u is the parent of v , v is called the child of u

Siblings: Vertices with the same parent are called siblings.

Ancestors: The ancestors of a vertex other than the root are the vertices in the path from the root to, this vertex, excluding the vertex itself and including the root.

Descendants: The descendants of a vertex v are those vertices that have v as an ancestor

Leaf: A vertex of a rooted tree is called a leaf if has no child.

Internal Vertices: Vertices that have children are called internal vertices. The root is an internal vertex unless it is the only vertex in the graph, in which case it is a leaf.

Subtree: If v is a vertex in a tree, the *subtree with v as its root* is the subgraph of the tree consisting of v and its descendants and all edges incident to these descendants.

8. Answer these questions about the rooted tree illustrated.

(a) Which vertex is the root? a

(b) Which vertices are internal? a, b, c

(c) Which vertices are leaves? f, g, e, d

(d) Which vertices are children of b ? f, g

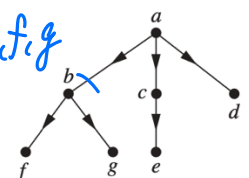
(e) Which vertex is the parent of b ? a

(f) Which vertices are siblings of d ? b, c

(g) Which vertices are ancestors of g ? a, b

(h) Which vertices are descendants of a ? b, c, d, e, f, g

(i) Find a subtree with b as its root.



9. Definition. An m -ary Tree:

A rooted tree is called an **m -ary tree** if every internal vertex has no more than m children. The tree is called a **full m -ary tree** if every internal vertex has exactly m children. An m -ary tree with $m = 2$ is called a **binary tree**.

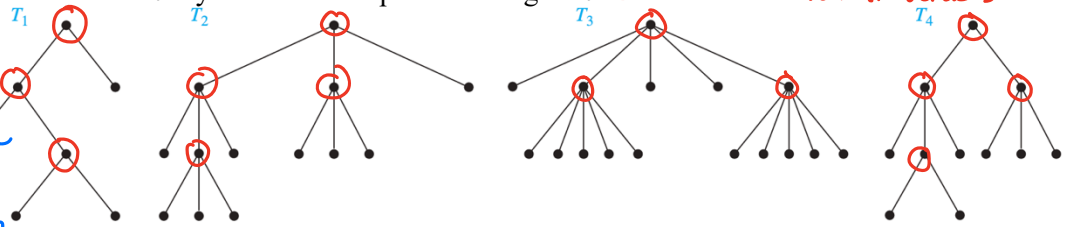
10. Examples. Check if each given tree is a full m -ary tree for some positive integer m . circle the internal vertices

T_1 : $m=2$, full binary tree

T_2 : $m=3$, full 3-ary tree

T_3 : $m=5$, full 5-ary tree

T_4 : $m=3$, 3-ary tree but not a full one



11. Theorem: A tree with n vertices has $n - 1$ edges. (i) and (ii) imply (iii)

Observation: (i) G is connected, (ii) G has no simple circuits, and (iii) G has $n - 1$ edges. Then we have if any two of the three are true, then the third one is true

12. Example. How many edges does a tree with 500 vertices have? $500 - 1 = 499$

13. Theorem: A full m -ary tree (a) with n vertices has $i = \frac{n-1}{m}$ internal vertices and $l = \frac{[(m-1)n+1]}{m}$ leaves, or (b) with i internal vertices contains $n = m \cdot i + 1$ vertices, and $l = (m-1)i + 1$ leaves, or (c) with l leaves has $n = \frac{ml-1}{m-1}$ vertices and $i = \frac{l-1}{m-1}$ internal vertices.

14. Example. How many vertices does a full 3-ary tree with 50 internal vertices have?

$3 \cdot 50 + 1 = 151$ vertices with $151 - 50 = 101$ leaves

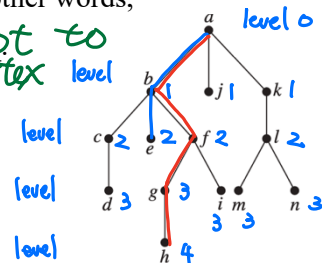
15. The Level of A Vertex and the Height of A Rooted Tree.

The **level of a vertex v** in a rooted tree is the length of the unique path from the root to this vertex. The level of the root is defined to be zero.

The **height of a rooted tree** is the maximum of the levels of vertices. In other words, the height of a rooted tree is the length of the longest path from the root to this vertex.

16. Example. Find the level of each vertex in the given rooted tree. What is the height of this tree?

the height is 4

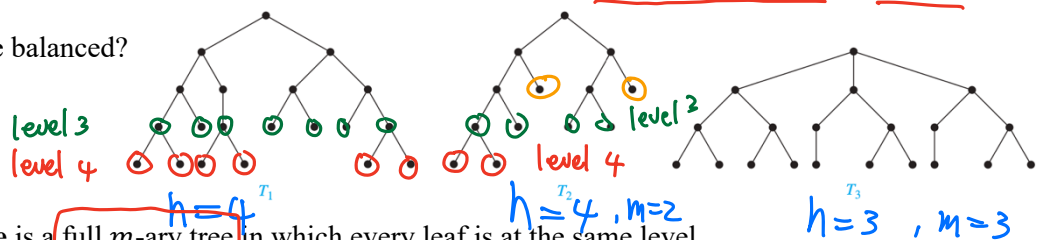


17. Definition. **Balanced m -ary Trees**: A rooted m -ary tree of height h is balanced if all leaves are at levels h or $h - 1$.

18. Which of the given rooted trees are balanced?

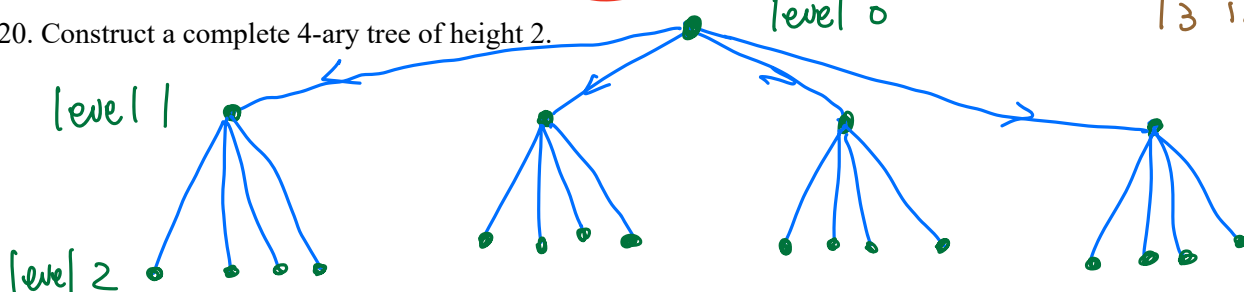
T_1 is a balanced 2-ary tree

T_2 is NOT a balanced binary tree



19. Definition. A **complete m -ary tree** is a full m -ary tree in which every leaf is at the same level.

20. Construct a complete 4-ary tree of height 2.



T_3 is a balanced 3-ary tree